

Jee Hoon Kim

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EDUCATION

Ph.D. (Sep 2024 - Current)

- **The University of Texas at Austin**, Fariborz Maseeh Department of Civil, Architectural, and Environmental Engineering
- Advisor: Prof. Christopher Rausch
- Research topic: Planning and Execution of field robotics for deconstruction projects

M.S. (Feb 2024)

- **Yonsei University**, Civil and Environmental Engineering
- GPA: 3.96 / 4.3
- Advisor: Prof. Hyoungkwan Kim
- Thesis title: "Automating Scaffold Safety Management through 3D Semantic Segmentation and Point Clouds to Image Transformation."

B.S (Feb 2022)

- **Yonsei University**, Civil and Environmental Engineering
 - GPA: 3.44 / 4.3
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WORK & RESEARCH EXPERIENCE

[Work Experience]

Amazon Fulfillment Technologies & Robotics, Seattle, WA, USA

Tech Deployment, BIM co-op (July 2025 – Dec 2025)

- Developed a project-agnostic deep learning-based point cloud segmentation software for tech deploy readiness
- Developed a layout converter software for robotic layout printing solutions
- Participated in few more other projects regarding process optimization/automation

MOHO Co., Republic of Korea

Research Engineer (Jun 2024 – Current)

- R&D for digital transformation of defect inspection/prediction of ceramic dishware products

ITHETTO Co., Republic of Korea

Research Engineer (Mar 2023 – Dec 2023)

- Developed a backpack shaped mobile scanning device for indoor construction sites
- Developed a structural inspection algorithm

[Research Experience]

Digital Harvest Lab, The University of Texas at Austin

Graduate Research Assistant (Sep 2024 – Current)

Advanced Infrastructure Management Group, Yonsei University

Post-graduate Researcher (Mar 2024 – Jun 2024)

Graduate Research Assistant (Mar 2022 – Feb 2024)

Undergraduate Intern (Feb 2021 – Feb 2022)

PUBLICATIONS

[Journal papers]

- **Jeehoon Kim** and Christopher Rausch "IFC-based Robotic Deconstruction Task and Motion Planning for Material Recovery in End-of-Life Buildings." *Under Review in Automation in Construction*
 - **Jeehoon Kim**, Juhyeon Kim, Nahye Koo, and Hyoungkwan Kim "Automating scaffold safety inspections using semantic analysis of 3D point clouds." *Automation in Construction* 166 (2024): 105603, doi: <https://doi.org/10.1016/j.autcon.2024.105603>.
 - Sunwoong Paik, **Jeehoon Kim**, Yulin Lian, Juhyeon Kim, Hyoungkwan Kim "Safety Analysis of Scaffold Joints through Image Generation from Point Cloud Data." *Journal of Construction Engineering and Management* <https://doi.org/10.1061/JCEMD4.COENG-18112>.
 - Yohan Kim, **Jeehoon Kim**, Juhyeon Kim, and Hyoungkwan Kim "Superpoint Transformer-based Bridge Component Recognition using UAV-mounted LiDAR and Synthetic Point Cloud Generation." *Journal of Computing in Civil Engineering* (2025). <https://doi.org/10.1061/JCCEE5.CPENG-6383>.
 - Juhyeon Kim, **Jeehoon Kim**, Yulin Lian, and Hyoungkwan Kim "Point cloud-based defect inspection with multi-
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sensor fusion and deep learning for advancing building construction quality.” *Journal of Computing in Civil Engineering* (2025). <https://doi.org/10.1061/JCCEE5.CPENG-6060>.

- Juhyeon Kim, **Jeehoon Kim**, Sunwoong Paik, and Hyoungkwan Kim “Point cloud registration considering safety nets during scaffold installation using sensor fusion and deep learning.” *Automation in Construction* 159 (2024): 105277, doi: <https://doi.org/10.1016/j.autcon.2024.105277>.
- Juhyeon Kim, **Jeehoon Kim**, Yohan Kim, and Hyoungkwan Kim “3D reconstruction of large-scale scaffolds with synthetic data generation and an upsampling adversarial network.” *Automation in Construction* 156 (2023): 105108, doi: <https://doi.org/10.1016/j.autcon.2023.105108>.

[International conference papers]

- **Jeehoon Kim** and Christopher Rausch (2025). “BIM-Driven Robotic Disassembly for Resource Recovery in End-of-Life Buildings.” *42th International Symposium on Automation and Robotics in Construction (ISARC 2025)*
- **Jeehoon Kim**, Sunwoong Paik, Yulin Lian, Juhyeon Kim, and Hyoungkwan Kim (2023). “Transformation of Point Clouds to Images for Safety Analysis of Scaffold Joints.” *40th International Symposium on Automation and Robotics in Construction (ISARC 2023)*, Chennai, India, 365-370.
- **Jeehoon Kim**, Juhyeon Kim, Nahye Koo, and Hyoungkwan Kim (2022). “Automated Checking of Scaffold Safety Regulations using Multi-Class 3D Segmentation.” *39th International Symposium on Automation and Robotics in Construction (ISARC 2022)*, Bogota, Columbia, 115-121.
- Juhyeon Kim, Duho Chung, **Jeehoon Kim**, and Hyoungkwan Kim (2025). “Integrated Framework for Scaffold Monitoring: From Point Cloud Acquisition to Safety Verification.” *42th International Symposium on Automation and Robotics in Construction (ISARC 2025)*
- Juhyeon Kim, **Jeehoon Kim**, Yulin Lian, and Hyoungkwan Kim (2024). “Automated Defect Inspection in Building Construction with Multi-Sensor Fusion and Deep Learning.” *41th International Symposium on Automation and Robotics in Construction (ISARC 2024)*, Lille, France, 1097-1103.
- Juhyeon Kim, **Jeehoon Kim**, Sunwoong Paik, and Hyoungkwan Kim (2023). “Sensor Fusion-based Point Clouds Registration Considering Safety Net During Scaffold Installation.” *ASCE International Conference on Computing in Civil Engineering (13CE 2023)*, Oregon, United States.
- Juhyeon Kim, **Jeehoon Kim**, Yohan Kim, and Hyoungkwan Kim (2022). “Large-scale scaffold point cloud data analysis framework using synthetic data augmentation and an upsampling adversarial network.” *22nd International Conference on Construction Applications of Virtual Reality (CONVR 2022)*, Seoul, Republic of Korea, 127-133.
- **Jeehoon Kim**, Minkyu Koo, Hongjo Kim, and Hyoungkwan Kim (2023). “Labeled Image Generation Using a Point Cloud-to-Image Transformation Algorithm.” *Korean Society of Civil Engineers 2023 convention conference and civil expo*, Yeosu, Republic of Korea.
- Yohan Kim, **Jeehoon Kim**, Juhyeon Kim, and Hyoungkwan Kim (2023). “3D Semantic Segmentation of Road Bridge Components using an Unmanned Aerial Vehicle Equipped with 3D Mobile LiDAR.” *Proceedings of KICEM Annual Conference 2023*, Gosung, Republic of Korea.

PATENTS & SOFTWARE (KR)

[Patents]

- “Method for modeling using product parameters and system therefor.” H. Lee, S. Jang, **J. Kim** (KR Patent 10-2025-0059702, 2025).
- (Applied) Title: “Labeled image data generation system using point cloud to image transformation.” H. Kim, **J. Kim**, KR Patent (2023).
- (Applied) Title: “Construction site temporary structure 3D modeling system and modeling method using synthetic data and generative adversarial neural network.” H. Kim, J. Kim, **J. Kim**, S. Paik, KR Patent (2023).
- (Applied) Title: “Safety inspection system for construction site temporary structures using semantic segmentation of point cloud data.” H. Kim, **J. Kim**, J. Kim, KR Patent (2022).

[Software]

- Title: “Point cloud to image transformation and image to point cloud reprojection module.” H. Kim, **J. Kim**, KR Software (2023).
- Title: “Temporary structure joint installation status automatic analysis module.” H. Kim, S. Paik, **J. Kim**, L. Yulin, KR Software (2022).
- Title: “Automated regulation checking algorithm for scaffolds.” H. Kim, **J. Kim**, KR Software (2022).

CERTIFICATES & SKILLS

Certificates

- 10-hour OSHA outreach for the construction industry (USWS45221738), *U.S. Department of Labor*
- Remote Pilot (Drone) License – part 107, *Federal Aviation Administration*.

Skills

- Skills: 3D Scanning (TLS, MLS, Photogrammetry), 3D Deep Learning, Sensors (Faro, intel RealSense, Azure Kinect, Velodyne, Ouster), Robots (Articulated robot, UGV, Drone), BIM (Revit, Navisworks)

TEACHING EXPERIENCE

Teaching Assistant , Construction Automation, Yonsei University	(Sep 2023 – Dec 2023)
Teaching Assistant , Construction Automation, Yonsei University	(Sep 2022 – Dec 2022)
